Clustering concepts

Module 4 - Lesson 1

Overview

Clustering finds groups in unlabeled data. K-means partitions data into k clusters by proximity; evaluating cluster quality is part of the workflow.

Key Concepts

- K-means assigns points to the nearest centroid and iteratively refines centroids.
- Choose k with the elbow method or silhouette score rather than guessing.
- Scale features before clustering so distances are meaningful.
- The inertia (within-cluster distance) decreases as k grows; the elbow shows the trade-off.
- Clusters reflect the chosen features and scale; they are descriptive, not causal.
- Visualize clusters with PCA or t-SNE for inspection.

Example

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

X_scaled = StandardScaler().fit_transform(X)
model = KMeans(n_clusters=4, n_init=10, random_state=42)
labels = model.fit_predict(X_scaled)
print(model.inertia_)
```

Summary

Clustering surfaces structure in unlabeled data. Pick k deliberately with the elbow or silhouette, scale features first, and treat clusters as descriptive groupings.

Practice Checklist

- Run K-means for several values of k and plot the elbow curve.
- Scale the data and re-run; describe how results change.
- Silhouette-score the clusters and pick the best k.